

# Leibniz Integral Rule

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## 0 The Leibniz Integral Rule

**Theorem 0.1** (Leibniz Integral Rule – Riemann Integral). Suppose  $I \subseteq \mathbb{R}$  is an open interval and  $f : [a, b] \times I \rightarrow \mathbb{R}$ ,  $(x, t) \mapsto f(x, t)$  such that  $f$  and  $\frac{\partial f}{\partial t}$  are continuous on  $[a, b] \times I$ . Define  $S : I \rightarrow \mathbb{R}$  as follows:

$$S(t) = \int_a^b f(x, t) dx$$

Then  $S$  is differentiable on  $I$  and:

$$S'(t) = \int_a^b \frac{\partial f}{\partial t}(x, t) dx$$

**Proof.** Suppose  $t_0 \in I$  and  $\varepsilon > 0$ . Since  $I$  is an open interval, there exists  $\delta_0 > 0$  such that  $[t_0 - \delta_0, t_0 + \delta_0] \subseteq I$ . Since  $\frac{\partial f}{\partial t}$  is continuous on  $[a, b] \times I$ , by the **Heine-Cantor theorem**, it is uniformly continuous on  $[a, b] \times [t_0 - \delta_0, t_0 + \delta_0]$ . Thus there exists  $0 < \delta < \delta_0$  such that:

$$\left| \frac{\partial f}{\partial t}(x, s) - \frac{\partial f}{\partial t}(x, t_0) \right| < \frac{\varepsilon}{b-a} \text{ for all } x \in [a, b] \text{ and all } s \in I, |s - t_0| < \delta$$

Now suppose  $t \in I$ ,  $|t - t_0| < \delta$  and  $x \in [a, b]$ . By **Lagrange's mean value theorem**, there exists  $\xi$  (possibly depending on  $x$ ) between  $t_0$  and  $t$  such that  $\frac{f(x, t) - f(x, t_0)}{t - t_0} = \frac{\partial f}{\partial t}(x, \xi)$ . This yields:

$$\left| \frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right| = \left| \frac{\partial f}{\partial t}(x, \xi) - \frac{\partial f}{\partial t}(x, t_0) \right| < \frac{\varepsilon}{b-a}$$

Note that the above inequality holds for all  $x \in [a, b]$ , even though  $\xi$  may depend on  $x$ . We also have:

$$\begin{aligned} \frac{S(t) - S(t_0)}{t - t_0} - \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx &= \frac{1}{t - t_0} \left( \int_a^b f(x, t) dx - \int_a^b f(x, t_0) dx \right) - \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx \\ &= \int_a^b \left( \frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right) dx \end{aligned}$$

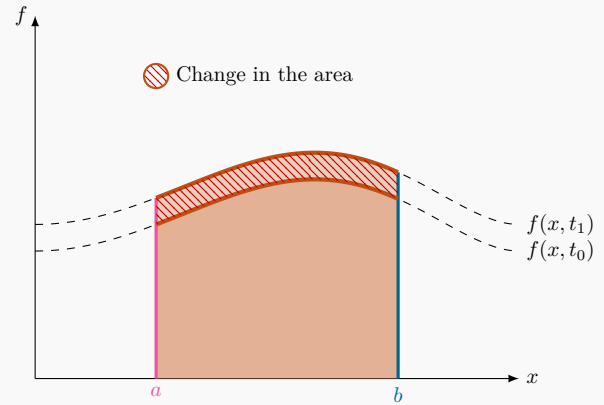
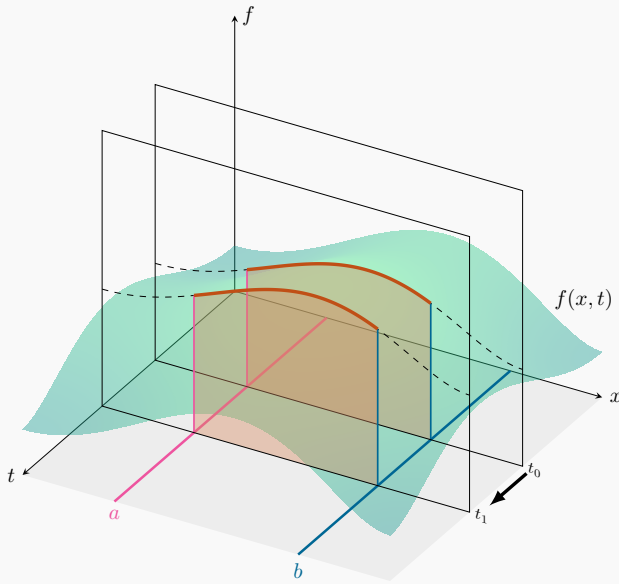
This yields:

$$\left| \frac{S(t) - S(t_0)}{t - t_0} - \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx \right| \leq \int_a^b \left| \frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right| dx < \int_a^b \frac{\varepsilon}{b-a} dx = (b-a) \frac{\varepsilon}{b-a} = \varepsilon$$

Thus  $S$  is differentiable at  $t_0$  and  $S'(t_0) = \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx$ . 🧐

## The Leibniz Integral Rule

$$\frac{d}{dt} \left( \int_a^b f(x, t) dx \right) = \int_a^b \frac{\partial f}{\partial t}(x, t) dx$$



As  $t$  moves from  $t_0$  to  $t_1$ , the shaded area changes from  $\int_a^b f(x, t_0) dx$  to  $\int_a^b f(x, t_1) dx$ .

Dividing this by  $t_1 - t_0$  and taking the limit as  $t_1 - t_0$  becomes small, we get  $\int_a^b \frac{\partial f}{\partial t}(x, t) dx$ .

**Theorem 0.2** (Leibniz Integral Rule – Improper Riemann Integral). Suppose  $I \subseteq \mathbb{R}$  is an open interval and  $f : [a, \infty) \times I \rightarrow \mathbb{R}$ ,  $(x, t) \mapsto f(x, t)$  such that:

- $f$  is continuous on  $[a, \infty) \times I$  improperly Riemann integrable with respect to  $x$  on  $[a, \infty)$  for each  $t \in I$ .
- $\frac{\partial f}{\partial t}$  is continuous on  $[a, \infty) \times I$  and improperly Riemann integrable with respect to  $x$  on  $[a, \infty)$ , uniformly in  $t \in I$ . In other words, for any  $\varepsilon > 0$ , there exists  $M > a$  (independent of  $t$ ) such that for all  $y > M$  and all  $t \in I$ , we have  $|\int_a^y f(x, t) dx| < \varepsilon$ .

Define  $S : I \rightarrow \mathbb{R}$  as follows:

$$S(t) = \int_a^\infty f(x, t) dx$$

Then  $S$  is differentiable on  $I$  and:

$$S'(t) = \int_a^\infty \frac{\partial f}{\partial t}(x, t) dx$$

**Proof.** Suppose  $(b_n)_{n=1}^\infty$  is a sequence in  $(a, \infty)$  and  $\lim_{n \rightarrow \infty} b_n = \infty$ . For each  $n \in \mathbb{N}$ , define  $S_n : I \rightarrow \mathbb{R}$  as follows:

$$S_n(t) = \int_a^{b_n} f(x, t) dx$$

By **Theorem 0.1**, for each  $n \in \mathbb{N}$ , we have  $S'_n(t) = \int_a^{b_n} \frac{\partial f}{\partial t}(x, t) dx$ . Since  $\frac{\partial f}{\partial t}(x, t)$  is improperly Riemann integrable with respect to  $x$  on  $[a, \infty)$ , uniformly in  $t$ , we have:

$$\lim_{n \rightarrow \infty} S'_n(t) = \int_a^\infty \frac{\partial f}{\partial t}(x, t) dx$$

**FINISH THE PROOF!!!**



The Leibniz integral rule can be applied to known integrals to derive related formulas:

**Example.**

$$\begin{aligned} \int_0^\infty \frac{1}{x^2 + a^2} dx &= \frac{\pi}{2a} & \xrightarrow{\frac{d}{da}} & \int_0^\infty \frac{1}{(x^2 + a^2)^2} dx = \frac{\pi}{4a^3} \\ & & \xrightarrow{\frac{d}{da}} & \int_0^\infty \frac{1}{(x^2 + a^2)^3} dx = \frac{3\pi}{16a^5} \\ & & \vdots & \\ & & \xrightarrow{\frac{d}{da}} & \int_0^\infty \frac{1}{(x^2 + a^2)^n} dx = \binom{2n-2}{n-1} \frac{\pi}{(2a)^{2n-1}} \end{aligned}$$

**Example.**

$$\begin{aligned} \int_0^\infty e^{-ax} dx &= \frac{1}{a} & \xrightarrow{\frac{d}{da}} & \int_0^\infty x e^{-ax} dx = \frac{1}{a^2} \\ & & \xrightarrow{\frac{d}{da}} & \int_0^\infty x^2 e^{-ax} dx = \frac{2}{a^3} \\ & & \vdots & \\ & & \xrightarrow{\frac{d}{da}} & \int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}} \end{aligned}$$

The Leibniz integral rule can also be used to solve certain integrals, by differentiating them first:

**Example.**

$$S(t) = \int_0^1 \frac{x^t - 1}{\log(x)} dx \quad (t > -1)$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^1 x^t dx = \frac{1}{t+1}$$

Integrating yields:

$$S(t) = \log(t+1) + C$$

Where  $C$  is a constant. Note that the integrand is zero when  $t = 0$ , so  $S(0) = 0$  and so  $C = 0$ . Thus:

$$\int_0^1 \frac{x^t - 1}{\log(x)} dx = \log(t+1)$$

**Example (Dirichlet integral).**

$$S(t) = \int_0^\infty e^{-tx} \frac{\sin(x)}{x} dx \quad (t > 0)$$

Differentiating under the integral, we get:

$$S'(t) = - \int_0^\infty e^{-tx} \sin(x) dx = - \frac{1}{t^2 + 1}$$

Integrating yields:

$$S(t) = -\arctan(t) + C$$

Where  $C$  is a constant. Note that the integrand tends to zero as  $t \rightarrow \infty$ , so  $\lim_{t \rightarrow \infty} S(t) = 0$  and so  $C = \frac{\pi}{2}$ . Thus:

$$\int_0^\infty e^{-tx} \frac{\sin(x)}{x} dx = \frac{\pi}{2} - \arctan(t)$$

Taking the limit as  $t \rightarrow 0^+$  yields:

$$\int_0^\infty \frac{\sin(x)}{x} dx = \frac{\pi}{2}$$

The following integral was solved in 1844 by [Joseph Serret](#) (1819 – 1885).

**Example (Serret's Integral).** Consider the following integral:

$$I = \int_0^1 \frac{\log(1+x)}{1+x^2} dx$$

To evaluate this, we first define  $S(t)$  as follows:

$$S(t) = \int_0^1 \frac{\log(1+tx)}{1+x^2} dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^1 \frac{x}{(1+tx)(1+x^2)} dx = -\frac{\log(1+t)}{1+t^2} + \frac{\pi t}{4(1+t^2)} + \frac{\log(2)}{2(1+t^2)}$$

Note that  $S(0) = 0$  and  $S(1) = I$ . Thus, we have:

$$\begin{aligned} I = S(1) - S(0) &= \int_0^1 S'(t) dt \\ &= -\int_0^1 \frac{\log(1+t)}{1+t^2} dt + \frac{\pi}{4} \int_0^1 \frac{t}{1+t^2} dt + \frac{\log(2)}{2} \int_0^1 \frac{1}{1+t^2} dt \\ &= -\int_0^1 \frac{\log(1+t)}{1+t^2} dt + \frac{\pi}{4} \left( \frac{\log(2)}{2} \right) + \frac{\log(2)}{2} \left( \frac{\pi}{4} \right) \\ &= -I + \frac{\pi}{4} \log(2) \end{aligned}$$

Rearranging yields  $2I = \frac{\pi}{4} \log(2)$ . Thus:

$$\int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi}{8} \log(2)$$

**Example.**

$$S(t) = \int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}} dx \quad (t \geq -1)$$

Differentiating under the integral (and assuming  $-1 \leq t \leq 1$  for now), we get:

$$S'(t) = \int_0^1 \frac{1}{(1+tx)\sqrt{1-x^2}} dx = \frac{\arccos(t)}{\sqrt{1-t^2}}$$

Integrating yields:

$$S(t) = -\frac{\arccos(t)^2}{2} + C$$

Where  $C$  is a constant. Note that the integrand is zero when  $t = 0$ , so  $S(0) = 0$  and so  $C = \frac{\pi^2}{8}$ . Thus:

$$\int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}} dx = \frac{\pi^2}{8} - \frac{\arccos(t)^2}{2}$$

If  $t > 1$ , we instead have:

$$S'(t) = \frac{\operatorname{arccosh}(t)}{\sqrt{t^2-1}}$$

This time, integrating yields:

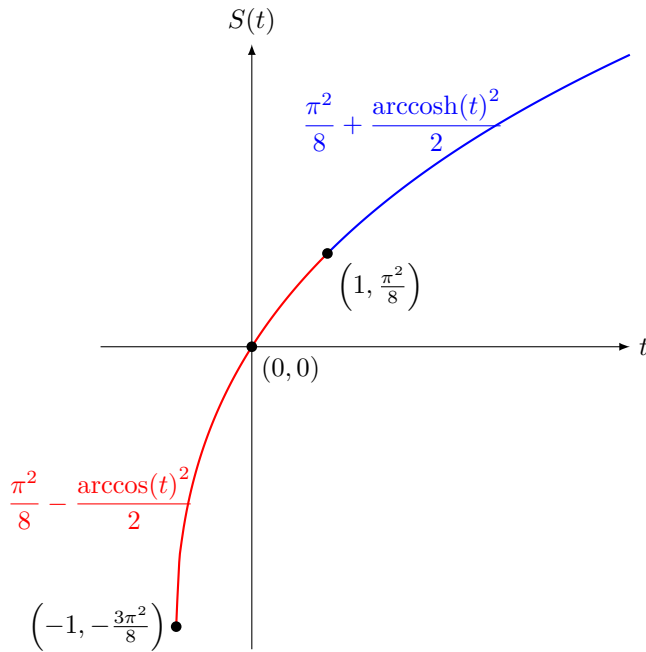
$$S(t) = \frac{\operatorname{arccosh}(t)^2}{2} + K$$

Where  $K$  is a constant. As we have already shown above, we have  $S(1) = \frac{\pi^2}{8}$  and so  $K = \frac{\pi^2}{8}$ . Thus:

$$\int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}} dx = \begin{cases} \frac{\pi^2}{8} - \frac{\arccos(t)^2}{2} & -1 \leq t \leq 1 \\ \frac{\pi^2}{8} + \frac{\operatorname{arccosh}(t)^2}{2} & t > 1 \end{cases}$$

Note that the function on the right, although defined piecewise, is analytic at every  $t > -1$  (including  $t = 1$ ).

$$\sum_{n=1}^{\infty} \frac{(-2)^{n+1}((n-1)!)^2}{(2n)!} (x-1)^n = \begin{cases} -\arccos(x)^2 & -1 \leq x \leq 1 \\ \operatorname{arccosh}(x)^2 & 1 \leq x \leq 3 \end{cases}$$



**Example.**

$$S(t) = \int_0^{\frac{\pi}{2}} \frac{\arctan(t \tan(x))}{\tan(x)} dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^{\frac{\pi}{2}} \frac{1}{1 + t^2 \tan^2(x)} dx = \frac{\pi}{2(t+1)}$$

Integrating yields:

$$S(t) = \frac{\pi}{2} \log(t+1) + C$$

Where  $C$  is a constant. Note that the integrand is zero when  $t = 0$ , so  $S(0) = 0$  and so  $C = 0$ . Thus:

$$\int_0^{\frac{\pi}{2}} \frac{\arctan(t \tan(x))}{\tan(x)} dx = \frac{\pi}{2} \log(t+1)$$

In particular, setting  $t = 1$  yields:

$$\int_0^{\frac{\pi}{2}} \frac{x}{\tan(x)} dx = \frac{\pi}{2} \log(2)$$

**Example.**

$$S(t) = \int_0^{\infty} e^{-x^2} \cos(tx) dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^{\infty} -xe^{-x^2} \sin(tx) dx$$

Integrating by parts yields:

$$S'(t) = \left[ -\frac{1}{2} e^{-x^2} \sin(tx) \right]_{x=0}^{x=\infty} - \int_0^{\infty} -\frac{t}{2} e^{-x^2} \cos(tx) dx = \frac{t}{2} \int_0^{\infty} e^{-x^2} \cos(tx) dx$$

Thus  $S'(t) = \frac{t}{2} S(t)$ . This is a linear first-order ODE in  $S(t)$ , whose solution is  $S(t) = C e^{-\frac{t^2}{4}}$ . To find  $C$ , note that:

$$S(0) = \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Thus:

$$\int_0^{\infty} e^{-x^2} \cos(tx) dx = \frac{\sqrt{\pi}}{2} e^{-\frac{t^2}{4}}$$

**Example.** This integral was discovered by Laplace in 1810:

$$S(t) = \int_0^{\infty} \frac{\cos(tx)}{x^2 + a^2} dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^\infty -\frac{x \sin(tx)}{x^2 + a^2} dx \quad (\text{Convergent for } t \neq 0)$$

Differentiating again, we get:

$$S''(t) = \int_0^\infty -\frac{x^2 \cos(tx)}{x^2 + a^2} dx \quad (\text{Divergent})$$

This is a problem, as the integral on the right does not converge. Instead, we can integrate the expression for  $S(t)$  by parts:

$$S(t) = \left[ \frac{\sin(tx)}{t(x^2 + a^2)} \right]_{x=0}^{x=\infty} - \int_0^\infty -\frac{2x \sin(tx)}{t(x^2 + a^2)^2} dx = \frac{2}{t} \int_0^\infty \frac{x \sin(tx)}{(x^2 + a^2)^2} dx$$

Call the integral on the right  $K(t)$ . Now we can differentiate under the integral:

$$\begin{aligned} K'(t) &= \int_0^\infty \frac{x^2 \cos(tx)}{(x^2 + a^2)^2} dx \\ &= \int_0^\infty \frac{\cos(tx)}{x^2 + a^2} dx - a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \\ &= S(t) - a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \end{aligned}$$

Since  $K(t) = \frac{t}{2}S(t)$ , we can rewrite the above equation as follows:

$$\begin{aligned} \frac{t}{2}S'(t) + \frac{1}{2}S(t) &= S(t) - a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \\ \frac{t}{2}S'(t) - \frac{1}{2}S(t) &= -a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \end{aligned}$$

Differentiating under the integral again, we get:

$$\frac{t}{2}S''(t) + \frac{1}{2}S'(t) - \frac{1}{2}S'(t) = a^2 \int_0^\infty \frac{x \sin(tx)}{(x^2 + a^2)^2} dx$$

The integral on the right is simply  $K(t) = \frac{t}{2}S(t)$ . This yields:

$$\frac{t}{2}S''(t) = a^2 \frac{t}{2}S(t)$$

Thus  $S''(t) = a^2 S(t)$ . This is a linear second-order ODE in  $S(t)$  with constant coefficients, whose solution is  $S(t) = C_1 e^{at} + C_2 e^{-at}$ . To find  $C_1$  and  $C_2$ , we need two equations. First, we have:

$$S(0) = \int_0^\infty \frac{1}{x^2 + a^2} dx = \frac{\pi}{2a}$$

Thus  $C_1 + C_2 = \frac{\pi}{2a}$ . Second, we have:

$$\lim_{t \rightarrow \infty} S(t) = \lim_{t \rightarrow \infty} \frac{2}{t} \int_0^\infty \frac{x \sin(tx)}{(x^2 + a^2)^2} dx = 0$$

This can also be seen by applying the [Riemann-Lebesgue lemma](#) directly to  $S(t)$ . Thus  $C_1 = 0$ , and so  $C_2 = \frac{\pi}{2a}$ . This yields:

$$S(t) = \frac{\pi}{2a} e^{-at} \quad (t \geq 0)$$

However, this is only valid for  $t \geq 0$ , as  $S(t)$  is not differentiable at  $t = 0$ . To handle the case  $t < 0$ , we can use the fact that  $S(t)$  is an even function of  $t$  (or alternatively, use the values  $S(0) = \frac{\pi}{2a}$  and  $\lim_{t \rightarrow -\infty} S(t) = 0$  to solve for  $C_1$  and  $C_2$ ). Thus, we have:

$$S(t) = \frac{\pi}{2a} e^{-a|t|}$$

**Example.** Define  $f : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$  as follows:

$$f(x, t) = \begin{cases} \frac{xt^3}{(x^2 + t^2)^2} & (x, t) \neq (0, 0) \\ 0 & (x, t) = (0, 0) \end{cases}$$

Then for all  $t \in \mathbb{R}$ , we have  $S(t) = \int_0^1 f(x, t) dx = \frac{t}{2(t^2 + 1)}$ . Thus  $S'(t) = \frac{1-t^2}{2(t^2 + 1)^2}$ , and so  $S'(0) = \frac{1}{2}$ . However,

$$\frac{\partial f}{\partial t} = \begin{cases} \frac{xt^2(3x^2 - t^2)}{(x^2 + t^2)^3} & (x, t) \neq (0, 0) \\ 0 & (x, t) = (0, 0) \end{cases} \quad \therefore \int_0^1 \frac{\partial f}{\partial t}(x, 0) dx = \int_0^1 0 dx = 0$$

The problem here is that  $\frac{\partial f}{\partial t}$  is not continuous (as a function of two variables) at  $(0, 0)$ , as  $\frac{\partial f}{\partial t}(t, t) = \frac{1}{4t}$  (for  $t \neq 0$ ), which blows up as  $t \rightarrow 0^+$ .

# 1 Other versions

The Leibniz integral rule can be reformulated using the Lebesgue integral:

**Theorem 1.1** (Leibniz Integral Rule – Lebesgue Integral). Suppose  $(X, \mathcal{A}, \mu)$  is a measure space,  $I \subseteq \mathbb{R}$  is an open interval and  $f : X \times I \rightarrow \mathbb{R}$  satisfies the following:

1. The function  $x \mapsto f(x, t)$  is integrable for all  $t \in I$
2. The function  $f \mapsto f(x, t)$  is differentiable for all  $x \in X$
3. There is an integrable function  $g : X \rightarrow \mathbb{R}$  such that  $\left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x)$  for all  $x \in X$  and all  $t \in I$

Then the function  $x \mapsto \frac{\partial f}{\partial t}$  is integrable for all  $t \in U$ , and:

$$\frac{d}{dt} \left( \int_X f(x, t) d\mu(x) \right) = \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x)$$

**Proof.** Suppose  $(h_n)_{n=1}^\infty$  is a sequence in  $\mathbb{R}$  and  $\lim_{n \rightarrow \infty} h_n = 0$ . For each  $n \in \mathbb{N}$ , define  $g_n : X \rightarrow \mathbb{R}$  as follows:

$$g_n(x) = \frac{f(x, t + h_n) - f(x, t)}{h_n} - \frac{\partial f}{\partial t}(x, t)$$

Then  $\lim_{n \rightarrow \infty} g_n(x) = 0$  for all  $x \in X$ . By [Lagrange's mean value theorem](#), we have:

$$\frac{f(x, t + h_n) - f(x, t)}{h_n} = \frac{\partial f}{\partial t} \text{ for some } \tau_n \text{ between } t \text{ and } t + h_n$$

Thus for all  $n \in \mathbb{N}$  and all  $x \in X$ , we have:

$$|g_n(x)| = \left| \frac{\partial f}{\partial t} - \frac{\partial f}{\partial t}(x, t) \right| \leq \left| \frac{\partial f}{\partial t} \right| + \frac{\partial f}{\partial t}(x, t) \leq g(x) + g(x) = 2g(x)$$

By the [dominated convergence theorem](#), we have:

$$\frac{1}{h_n} \int_X (f(x, t + h_n) - f(x, t)) d\mu(x) - \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x) = \int_X g_n(x) d\mu(x) \rightarrow 0 \text{ as } n \rightarrow \infty$$

As  $n \rightarrow \infty$ , the first integral tends to  $\frac{d}{dt} \left( \int_X f(x, t) d\mu(x) \right)$ . Since this holds for every sequence  $(h_n)_{n=1}^\infty$  that converges to zero, we have:

$$\frac{d}{dt} \left( \int_X f(x, t) d\mu(x) \right) = \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x) \quad \text{😎}$$

The Leibniz integral rule can also be adapted to the case where the limits are not constants but functions of  $t$ :

**Theorem 1.2** (Leibniz Integral Rule – Variable Limits). Suppose  $I \subseteq \mathbb{R}$  is an open interval,  $a, b : I \rightarrow \mathbb{R}$  are differentiable on  $I$  such that  $a(t), b(t) \in [A, B]$  for all  $t \in I$ , and  $f : [a, b] \times I \rightarrow \mathbb{R}$ ,  $(x, t) \mapsto f(x, t)$  such that  $f$  and  $\frac{\partial f}{\partial t}$  are continuous on  $[A, B] \times I$ . Define  $S : I \rightarrow \mathbb{R}$  as follows:

$$S(t) = \int_{a(t)}^{b(t)} f(x, t) dx$$

Then  $S$  is differentiable on  $I$  and:

$$S'(t) = b'(t)f(b(t), t) - a'(t)f(a(t), t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) dx$$

**Proof.** Define  $G : [A, B] \times [A, B] \times I \rightarrow \mathbb{R}$  as follows:

$$G(a, b, t) = \int_a^b f(x, t) dx$$

In other words,  $G$  is a function of three independent variables  $a, b, t$ . By the [fundamental theorem of calculus](#), we have:

$$\frac{\partial G(a, b, t)}{\partial b} = f(b, t) \qquad \frac{\partial G(a, b, t)}{\partial a} = -f(a, t)$$

By [Theorem 0.1](#), we also have:

$$\frac{\partial G(a, b, t)}{\partial t} = \int_a^b \frac{\partial f}{\partial t}(x, t) dx$$

Thus, by the multivariable chain rule, we have:

$$\begin{aligned}\frac{d}{dt}G(a(t), b(t), t) &= \frac{db(t)}{dt} \frac{\partial G}{\partial b}(a(t), b(t), t) + \frac{da(t)}{dt} \frac{\partial G}{\partial a}(a(t), b(t), t) + \frac{dt}{dt} \frac{\partial G}{\partial t}(a(t), b(t), t) \\ &= b'(t) \frac{\partial G}{\partial b}(a(t), b(t), t) + a'(t) \frac{\partial G}{\partial a}(a(t), b(t), t) + \frac{\partial G}{\partial t}(a(t), b(t), t) \\ &= b'(t)f(b(t), t) - a'(t)f(a(t), t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) dx\end{aligned}$$



Fix  $t_0 \in I$ .

$$\begin{aligned}S(t) &= \int_{a(t)}^{b(t)} f(x, t) dx \\ &= \int_{b(t_0)}^{b(t)} f(x, t) dx - \int_{a(t_0)}^{a(t)} f(x, t) dx + \int_{a(t_0)}^{b(t_0)} f(x, t) dx \\ &= \int_{b(t_0)}^{b(t)} f(x, t_0) dx - \int_{a(t_0)}^{a(t)} f(x, t_0) dx + \int_{a(t_0)}^{b(t_0)} f(x, t) dx \\ &\quad + \int_{b(t_0)}^{b(t)} (f(x, t) - f(x, t_0)) dx - \int_{a(t_0)}^{a(t)} (f(x, t) - f(x, t_0)) dx\end{aligned}$$

We first show that  $S'_1(t) = b'(t)f(b(t), t)$ . By the **fundamental theorem of calculus** and the **chain rule**, we have:

$$S'_1(t) = b'(t)f(b(t), t) \qquad S'_2(t) = a'(t)f(a(t), t)$$

By **Theorem 0.1**, we also have:

$$S'_3(t) = \int_{a(t_0)}^{b(t_0)} \frac{\partial f}{\partial t}(x, t) dx$$

Since  $b$  is differentiable at  $t_0$ , it is continuous at  $t_0$ , so  $\lim_{t \rightarrow t_0} |b(t) - b(t_0)| = 0$ . Since  $\frac{\partial f}{\partial t}$  is continuous at  $t_0$ , it is bounded in a neighborhood of  $t_0$ , i.e. there exists

$$\begin{aligned}\frac{S_4(t) - S_4(t_0)}{t - t_0} &= \frac{1}{t - t_0} \int_{b(t_0)}^{b(t)} (f(x, t) - f(x, t_0)) dx \\ &= \int_{b(t_0)}^{b(t)} \frac{f(x, t) - f(x, t_0)}{t - t_0} dx \\ &\leq |b(t) - b(t_0)| \sup_{\xi \in B} \int_{b(t_0)}^{b(t)} \frac{f(x, t) - f(x, t_0)}{t - t_0} dx\end{aligned}$$

**Example** (Variant of Serret's integral).

$$S(t) = \int_0^t \frac{\log(1+tx)}{1+x^2} dx$$

Differentiating under the integral, we get:

$$\begin{aligned}S'(t) &= \frac{\log(1+t^2)}{1+t^2} - 0 + \int_0^t \frac{x}{(1+tx)(1+x^2)} dx \\ &= \frac{\log(1+t^2)}{1+t^2} - 0 - \frac{\log(1+t^2)}{1+t^2} + \frac{t \arctan(t)}{1+t^2} + \frac{\log(1+t^2)}{2(1+t^2)} \\ &= \frac{t \arctan(t)}{1+t^2} + \frac{\log(1+t^2)}{2(1+t^2)}\end{aligned}$$

Integrating the first term by parts yields:

$$\int \frac{t \arctan(t)}{1+t^2} dt = \frac{\arctan(t) \log(1+t^2)}{2} - \int \frac{\log(1+t^2)}{2(1+t^2)} dt$$

Thus the second term cancels out, and so:

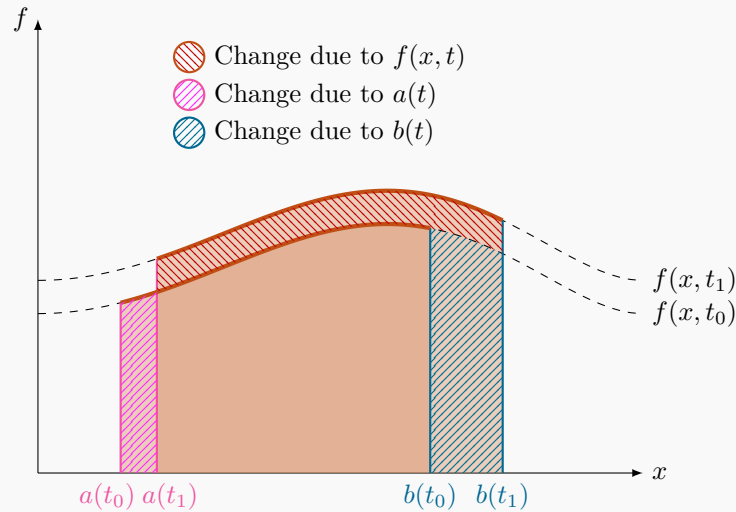
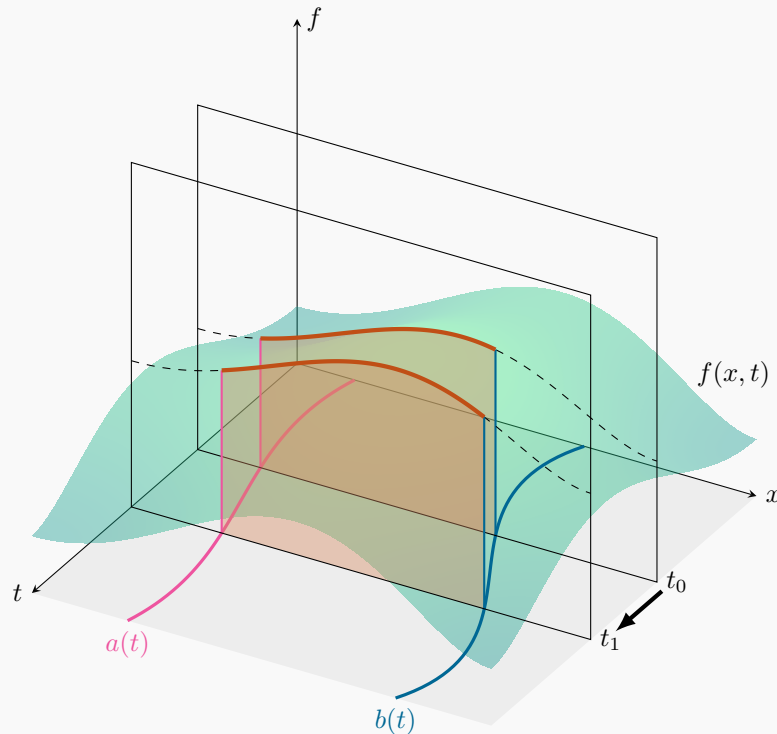
$$S(t) = \frac{\arctan(t) \log(1+t^2)}{2} + C$$

Where  $C$  is a constant. As with Serret's integral, the integrand vanishes when  $t = 0$ , so  $S(0) = 0$  and so  $C = 0$ . Thus:

$$\int_0^t \frac{\log(1+tx)}{1+x^2} dx = \frac{\arctan(t) \log(1+t^2)}{2}$$

## The Leibniz Integral Rule (Variable Limits)

$$\frac{d}{dt} \left( \int_{a(t)}^{b(t)} f(x, t) dx \right) = b'(t)f(b(t), t) - a'(t)f(a(t), t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) dx$$



We now use the Leibniz integral rule to prove the **fundamental theorem of algebra**:

**Theorem 1.3** (Fundamental Theorem of Algebra). Every polynomial  $p : \mathbb{C} \rightarrow \mathbb{C}$  has at least one root.

**Proof.** Suppose  $p(z) \neq 0$  for all  $z \in \mathbb{C}$ . Define  $I : [0, \infty) \rightarrow \mathbb{C}$  as follows:

$$I(r) = \int_0^{2\pi} \frac{1}{p(re^{i\theta})} d\theta$$

This is well-defined as  $p(z)$  is never zero, so  $\frac{1}{p(z)}$  is continuous on  $\mathbb{C}$ .

We first show that  $\lim_{r \rightarrow 0^+} I(r) = I(0)$ , i.e.  $I$  is continuous at 0. Since  $\frac{1}{p(z)}$  is continuous at 0, for any  $\varepsilon > 0$ , there

exists  $\delta > 0$  such that  $\left| \frac{1}{p(z)} - \frac{1}{p(0)} \right| < \frac{\varepsilon}{2\pi}$  for all  $z \in \mathbb{C}$ ,  $|z| < \delta$ . Thus for all  $0 \leq r < \delta$ , we have:

$$|I(r) - I(0)| = \left| \int_0^{2\pi} \left( \frac{1}{p(re^{i\theta})} - \frac{1}{p(0)} \right) d\theta \right| \leq \int_0^{2\pi} \left| \frac{1}{p(re^{i\theta})} - \frac{1}{p(0)} \right| d\theta < \int_0^{2\pi} \frac{\varepsilon}{2\pi} d\theta = \varepsilon$$

Thus  $\lim_{r \rightarrow 0^+} I(r) = I(0)$ .

We now show that  $\lim_{r \rightarrow \infty} I(r) = 0$ . Suppose the leading (highest-degree) term of  $p(z)$  is  $az^d$ . Then we have  $\lim_{r \rightarrow \infty} \frac{|p(re^{i\theta})|}{|re^{i\theta}|^d} = |a|$ , so there exists  $R > 0$  such that  $|p(z)| > \frac{|a|}{2} r^d$  for all  $r > R$ . Thus for all  $r > R$ , we have:

$$|I(r)| \leq \int_0^{2\pi} \frac{1}{|p(re^{i\theta})|} d\theta \leq \int_0^{2\pi} \frac{2}{|a|r^d} d\theta = \frac{4\pi}{|a|r^d}$$

Since the right side tends to zero as  $r \rightarrow \infty$ , we have  $\lim_{r \rightarrow \infty} I(r) = 0$ .

Note that:

$$\frac{\partial}{\partial \theta} \left( \frac{1}{p(re^{i\theta})} \right) = -ire^{i\theta} \frac{p'(re^{i\theta})}{p(re^{i\theta})^2} = ir \frac{\partial}{\partial r} \left( \frac{1}{p(re^{i\theta})} \right)$$

Thus, differentiating under the integral, we get:

$$I'(r) = \int_0^{2\pi} \frac{\partial}{\partial r} \left( \frac{1}{p(re^{i\theta})} \right) d\theta = \int_0^{2\pi} \frac{1}{ir} \frac{\partial}{\partial \theta} \left( \frac{1}{p(re^{i\theta})} \right) d\theta = \frac{1}{ir} \left( \frac{1}{p(re^{i\theta})} \right)_{\theta=0}^{\theta=2\pi} = \frac{1}{ir} \left( \frac{1}{p(r)} - \frac{1}{p(r)} \right) = 0$$

Thus  $I$  is constant on  $[0, \infty)$ . This is a contradiction, as  $\lim_{r \rightarrow \infty} I(r) = 0$  but  $\lim_{r \rightarrow 0^+} I(r) = \frac{2\pi}{p(0)} \neq 0$ . Thus  $p$  must have at least one root, i.e.  $p(z) = 0$  for some  $z \in \mathbb{C}$ . 🤖